

Feliks Victor Parningotan Samosir

Full-time Lecturer, Faculty of Artificial Intelligence and Data Science,
Universitas Pelita Harapan (UPH), Tangerang, Banten, Indonesia

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Nationality: Indonesian

Research Interests: Low-resource natural language processing, with a focus on Toba Batak and Indonesian; dependency parsing and linguistic typology; neural machine translation; sentiment analysis; text summarization; corpus construction and dataset quality; bias and fairness in Indonesian NLP resources.

EDUCATION

Master of Computer Science

Aug 2020 – May 2022

Universitas Kristen Maranatha, Bandung, West Java

Thesis area: BERT-based extractive summarization with clustering for scientific papers.

Bachelor of Education in Informatics Engineering

Aug 2011 – Jan 2018

Universitas Negeri Malang, Malang, East Java

WORK EXPERIENCE

Full-time Lecturer, Faculty of Artificial Intelligence and Data Science

Jan 2023 – Present

Universitas Pelita Harapan, Tangerang

Adjunct Lecturer

Apr 2022 – Dec 2022

Raffles Institute of Higher Education, Jakarta

Information Technology Support Specialist

Sep 2019 – Jan 2023

Sekolah Tinggi Alkitab Tiranus, Bandung

Product Development Staff

Mar 2018 – Jun 2019

Kapanlagi Youniverse, Malang

Computer Teacher

Feb 2015 – Jun 2017

Charis National Academy, Malang

TEACHING EXPERIENCE

Universitas Pelita Harapan

2023 – present

Natural Language Processing; Advanced Machine Learning; Introduction to Data Science; Data Science Application Development Studio; Database Systems; Object-Oriented Modeling; Introduction to Informatics and Computing.

Supervision: Mentor undergraduate research projects on Transformer-based question answering, text classification, text summarization, and bias and fairness analysis in Indonesian NLP datasets.

Raffles Institute of Higher Education

2022

Neural Networks Basics; Data Analysis; Statistics for AI; Algorithms; Discrete Mathematics; Databases; Software Specification and Modeling; Python; Introduction to Computational Thinking; Operating Systems and Computer Networks.

RESEARCH EXPERIENCE

Universal Dependencies Annotation Guidelines for Toba Batak with Initial Validation on a Small Corpus, PI, UPH Internal Research Fund (IRF)

Jul 2026 – Dec 2026 (ongoing)

Second stage of a long-term Toba Batak NLP programme, extending the raw corpus built under the 2025 grant. Drafting the first Universal Dependencies annotation guidelines for this language, focusing on predicate-initial word order and the western Austronesian voice system, and validating them through pilot annotation of 250 sentences with inter-annotator agreement measurement. Coordinating a consulting linguist and two annotators toward a journal article, the guideline document, and an openly released CoNLL-U corpus.

Developing Toba Batak Language Corpus, PI, UPH IRF*May 2025 – Dec 2025*

Designed the corpus schema and genre taxonomy for multi-domain Toba Batak and Indonesian texts covering religious, cultural, educational, and contemporary sources. Led the end-to-end digitisation pipeline, including document collection, scanning and OCR, and manual correction with native Toba Batak speakers, followed by cleaning, tokenisation, normalisation, and metadata annotation. Prepared the corpus and its documentation as a foundation for future local-language models and evaluation benchmarks.

Bias Analysis in Indonesian QA Datasets, PI, UPH IRF*May 2024 – Dec 2024*

Investigated bias and fairness issues in the Indonesian FACQA dataset, designed an experimental protocol using Transformer baselines and quantitative fairness metrics, conducted error analysis on how linguistic patterns and dataset design amplify or reduce bias, and proposed documentation and mitigation recommendations for Indonesian NLP resources.

Extractive Summarization for Mind Map Generation, PI, UPH IRF*May 2023 – May 2024*

Developed a prototype using Transformer-based extractive summarization and clustering to generate structured mind maps from scientific articles, evaluated with lecturers and students on informativeness, readability, and usefulness for learning, and integrated it into teaching activities.

PUBLICATIONS (ACADEMIC WORKS)

Selected from all publications. Full list available via Google Scholar.

1. F. V. P. Samosir, M. Diana, and K. Wiriyatama, “A Multi-Dimensional Framework for Bias Assessment in Indonesia Question-Answering Datasets,” *JOIV: International Journal on Informatics Visualization*, vol. 10, no. 3, p. 1376, May 2026, doi: 10.62527/joiv.10.3.4122.
2. F. V. P. Samosir and G. L. Tumanggor, “Integrated aspect extraction and sentiment classification for aspect-based sentiment analysis using fine-tuned IndoBERT on Indonesian e-commerce reviews,” *Journal of Soft Computing Exploration*, vol. 7, no. 1, pp. 147–156, Apr. 2026, doi: 10.52465/joscex.v7i1.26.
3. F. V. P. Samosir, J. Christian, J. M. Winarko, J. Tiong, V. Arianto, and M. K. Grady, “Personal Document Classification using Deep Learning Approach: a Comparative Study of CNN, MobileNetV2, and Vision Transformers,” in *Proc. 2025 10th Int. Conf. Informatics and Computing (ICIC)*, Lampung, Indonesia: IEEE, Oct. 2025, pp. 1–6, doi: 10.1109/ICIC68054.2025.11309527.
4. F. V. P. Samosir and S. Mendrofa, “IndoBerea: Evolving Semantic Search in Theological Context,” in *Proc. 2023 8th Int. Conf. Informatics and Computing (ICIC)*, Manado, Indonesia: IEEE, Dec. 2023, pp. 1–6, doi: 10.1109/ICIC60109.2023.10382053.
5. F. V. P. Samosir, H. Toba, and M. Ayub, “BESKlus: BERT Extractive Summarization with K-Means Clustering in Scientific Paper,” *Jurnal Teknik Informatika dan Sistem Informasi*, vol. 8, no. 1, pp. 202–217, Apr. 2022, doi: 10.28932/jutisi.v8i1.4474.

SKILLS

Programming and Frameworks: Python; PyTorch, TensorFlow, Hugging Face Transformers.

Libraries and Tools: pandas, NumPy, Matplotlib, scikit-learn; Git, Visual Studio Code, Google Colab, Gradio.

Research Methods: Corpus construction and annotation workflow design, dataset quality and bias analysis, Transformer fine-tuning, evaluation and benchmarking for low-resource languages.

Languages: Indonesian, Toba Batak, English.